A checked point in the actual weighted-Proj chart overlap

Formal proof packet

28 October 2026

For every field K and coefficient tuple, the new Lean theorem constructs a point of the actual weighted-Proj curve belonging to both its X- and Z-basic opens. No separability hypothesis is needed for this point-level statement. Pinned Lean 4.34.1 exited zero; the direct axiom audit lists only `propext`, `Classical.choice`, and `Quot.sound`.

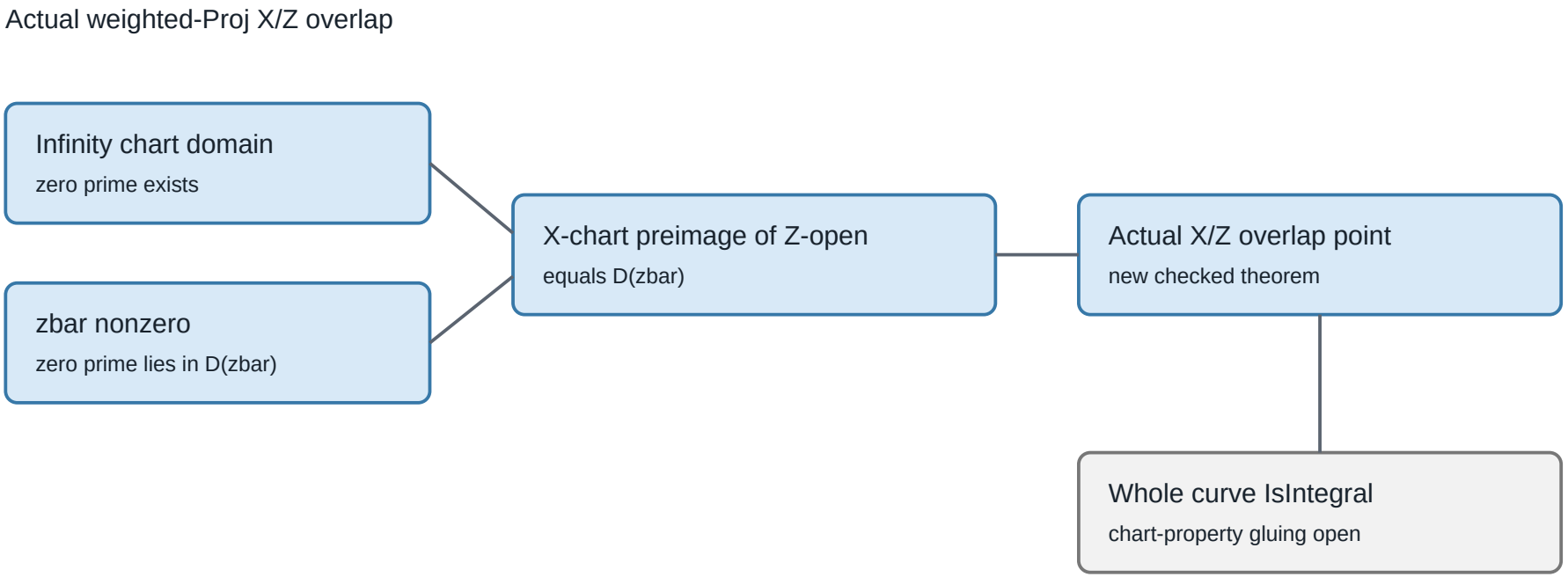
Witness and checked dependencies

The infinity-chart coordinate ring is a domain, so its zero prime exists. The previously checked $\bar{z} \neq 0$ lemma places that prime in $D(\bar{z})$. The existing checked cross-chart theorem identifies $D(\bar{z})$ with the inverse image of the actual weighted-Proj Z-open under the X-chart map. By definition, this map lands in the X-open. The source theorem combines these facts into an explicit point of the intersection.

Remaining boundary

The new theorem proves nonempty overlap only. A whole-curve `IsIntegral` instance still needs the two chart-domain properties transported across their scheme isomorphisms, then irreducibility and reducedness glued over the checked two-open cover. The affine chart domain currently requires a separable quintic. Principal divisors at all closed points, Picard-class transport, and compiled runtime semantics remain open. The Markdown and CSV give exact declarations and statuses. No DLP experiments or cost comparisons were run.

Proof dependencies



Blue = checked declarations. Gray = remaining proof. Arrows = dependencies, not measurements.

Blue nodes are checked; the gray node is a remaining proof obligation. Arrows are dependencies, not performance measurements.